

PRATUSH BRAHMA

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PROFESSIONAL SUMMARY

Wildlife biologist and environmental scientist with extensive field and analytical experience in ecological research, habitat assessment, wildlife surveys, and spatial data analysis across the United States and India. Proficient in quantitative data analysis, technical writing, and tools including R and QGIS. Experienced educator and program administrator with a strong record of leading teams, managing multi-course academic programs, and mentoring students at multiple levels. Demonstrated leadership in professional, academic, and competitive settings including national championship winning teams.

EDUCATION

Doctor of Philosophy, University of Florida, Gainesville, FL *Anticipated June 2025*

Dissertation: Understanding the Evolution of Animal Behavior

Integrated Master of Science, National Institute of Science Education and Research, India *June 2018*

Major: Cancer Genetics and Angiogenesis | Cumulative GPA: 3.35/4.00

STRENGTHS AND SPECIAL SKILLS

- o Extensive experience with Dynamic State Variable Models, Agent-based Models, and Population Viability Analysis
- o Proficient in R, Shell, LaTeX, RMarkdown; Analytical tools: ImageJ, QGIS, MEGA, PyMol, SnapGene
- o Wildlife handling experience: spiders, lizards, fish, and mice
- o Skilled in field data collection, microscopy, spatial mapping, and photography
- o Excellent written and oral communication skills
- o Experience in undergraduate teaching, program coordination, and public science outreach
- o Demonstrated ability to work in remote field sites across India and the USA

PROFESSIONAL EXPERIENCE

Program Director, Precollege Scholars Program *2024 to Present*

Center for Precollegiate Education and Training (CPET), University of Florida

- o Lead and oversee all aspects of the Precollege Scholars Program including program coordination, student engagement, and staff management
- o Manage and coordinate delivery of multiple precollege level courses simultaneously
- o Develop and review educational content, curriculum, and instructional materials
- o Serve in a senior leadership and administrative capacity overseeing full program operations

Program Coordinator, Precollegiate Scholars Program *Summer 2023*

Center for Precollegiate Education and Training (CPET), University of Florida

- o Wrote, reviewed, and developed educational content for online courses
- o Coordinated setup and functioning of over 10 courses simultaneously
- o Assisted with student enrollment and engagement

Camp Counselor and Science Instructor, Summer Camp *Summer 2023*

Center for Precollegiate Education and Training, University of Florida

- o Developed and delivered educational content including lectures and hands-on activities
- o Communicated effectively with fellow staff in a fast-paced environment
- o Provided instruction and management to children aligned with summer camp objectives

Doctoral Researcher, Invasive Species Management *August 2018 to Present*

Department of Biology, University of Florida

- o Created optimality models to obtain optimal trajectories of behavioral choices to maximize fitness under varying density conditions
- o Developed individual-based simulations to understand fitness consequences of behavioral adaptations
- o Conducted population viability analysis for invasive fish species
- o Engaged in invasive species handling and aquaculture

Doctoral Researcher, Wildlife Ecological Assessment

August 2022 to Present

Department of Entomology, University of Florida

- o Conducted fieldwork collecting over 50 jumping spiders in Gainesville, Florida
- o Designed and conducted behavioral experiments on the evolution of color vision
- o Engaged in animal handling and rearing of insects including spiders, crickets, and termites
- o Used statistical tools for data analysis and interpretation

Research Collaborator, Wildlife Ecological Assessment

May 2023 to March 2024

Azim Premji University, India

- o Conducted fieldwork collecting over 200 jumping spiders in Bengaluru, India
- o Designed and conducted behavioral experiments on color vision evolution
- o Mentored undergraduate students in scientific research and critical thinking

Teaching Assistant and Lab Technician

August 2018 to Present

Department of Biology, University of Florida

- o Wrote and developed educational content for laboratory and theoretical classes
- o Delivered lectures in Introductory Biology Labs and Genetics
- o Designed an interactive simulation tool for undergraduate evolutionary biology
- o Conducted research on native flora with undergraduate students

Masters Researcher, Cancer Angiogenesis

May 2016 to June 2018

National Institute of Science Education and Research, India

- o Conducted lab studies using mammalian cell lines, mice models, chicken embryos, and cancer tissues
- o Applied statistical tools for data analysis and interpretation

Student Intern, Field Ecology, Western Ghats

3 months, India

- o Identified and spatially mapped over 200 Rock Agamas using field identification techniques and GIS
- o Conducted count surveys and body measurements to record abundance and survival tracking
- o Used photography and GIS mapping to determine individual territories

Student Intern, Genetic Analysis, NISER

3 months, India

- o Extracted DNA and conducted PCR and RFLP analysis on over 100 gallbladder cancer samples

Student Intern, Phylogenetics, Indian Institute of Science

3 months, India

- o Conducted morphometric and genetic analyses to generate phylogenies of Microhylidae frogs using MEGA software

LEADERSHIP AND EXTRACURRICULAR

Team Captain, Local City Cricket Club

2021 to Present

- o Led the team to victory in the annual city cricket championship for four consecutive years

Club President and Team Captain, University of Florida Cricket Team

University of Florida

- o Served as Club President and Team Captain
- o Led the team to Runners-Up at the Collegiate Cricket Nationals in 2022, in the second year with the team

Treasurer and Team Captain, University of Florida Badminton Team

University of Florida

- o Served as Treasurer and Team Captain

- o Led the team to the National Championship in 2019, the first national title in UF Badminton history, in the first year with the team

OUTREACH EXPERIENCE

Florida Museum of Natural History, Gainesville, FL

Fall 2024

- o Delivered an arachnid awareness program using creative techniques and live spiders for the general public

iThink Biology, NPTEL Course, India

Spring 2024

- o Interviewed a public health minister for inclusion in an online course creating public health awareness for youth in India

PUBLICATIONS

- o Naik M., Brahma P., and Dixit M. (2018). Cost-effective ex ovo CAM assay protocol. *Methods and Protocols*, 1(2), 19.
- o Mukherjee B., Brahma P., et al. (2023). FRG1 in angiogenesis via FGF2-ERK/AKT. *FEBS Open Bio*, 13(5), 804-817.
- o [In Preparation] Brahma P., Lietzenmayer L., Menon Y., Colette S.M., Uma D., and Taylor L. (2024). The use of color in foraging in Lessert's Rainbow Spider (*Stenaelurillus lesserti*).
- o [In Preparation] Brahma P. and Colette S.M. (2024). Variation in sex-specific reproductive strategies in response to changes in population density.
- o [In Preparation] Lietzenmayer L., et al., Brahma P., and Taylor L. (2024). Does color vision improve jumping spiders' ability to find colored prey on colored backgrounds?

RELEVANT COURSEWORK

- o Wildlife Management Techniques | Quantitative Modeling | Animal and Plant Physiology | Evolutionary Biology
- o Wildlife Ecology and Behavior | Molecular Biology and Immunology | Biotechnology | Biostatistics and Bioinformatics